

P-Value: A p-value lower than 0.05 means it's statistically significant. The lower the p-value the greater its statistical significance.

ADF Test/Dickey-Fuller Test: Whether a time series dataset is stationary, checks the presence of a unit root. The more negative the value, the more strongly the rejection of the hypothesis at the unit root takes place.

Lag Order: Tell us the optimal number of lags which need to be included in a time series model to capture the structure.

Stationary: Data sequence whose statistical properties such as mean, variance and autocorrelation remain the same over time, lacks trends or any seasonal pattern. Desired for time series to be stationary.

Limited Autocorrelation: A variable's current value has little or no linear relationship with its own previous values.

Volatility Clustering - Financial phenomena by which high-volatility periods tend to follow high-volatility periods and low-volatility periods then to follow low-volatility periods, meaning price changes cluster together.

R² Value – Indicates how much of the variation of dependent variable is explained by the independent variable in a regression model. An independent variable is the presumed cause, variable that has been manipulated or controlled to study its effect. Dependent variable is the outcome measured, changes based on the independent variable.

Example: The effect of study time on test scores

Independent Variable – Time spent studying

Dependent Variable – Test scores (Depends on time spent)

Lurking Variables – Any other factors outside of IV and DV which impacts the study, creates a false perception of cause and effect.

R-Value – Correlation coefficient, measures the strength of the relationship between two variables. Ranges from -1 to 1. Coefficient of 1, shows a perfect positive correlation, coefficient of -1, shows a perfect negative correlation.

1. What the project set out to do

Your project analyzed **Bitcoin price and return dynamics** using classical time-series methods to answer two questions:

1. **Can the mean behavior of Bitcoin returns be modeled with linear time-series models?**
2. **How does volatility evolve over time, especially during market stress?**

This is a standard but rigorous entry point into **financial time-series inference**.

2. Main empirical findings (what the data showed)

Finding 1: Bitcoin returns are weakly predictable in the mean

- After differencing/log-returns, the series was approximately **mean-stationary**.
- ARIMA components captured only **limited autocorrelation**.
- Most predictive structure in the mean disappeared after basic filtering.

Interpretation:

Bitcoin behaves like many financial assets: **returns are close to a martingale**, but volatility is not.

Finding 2: Strong volatility clustering

You found clear evidence that:

- Large price changes cluster in time,
- Periods of calm alternate with periods of extreme turbulence.

This was supported by:

- Significant ARCH effects,
- Persistent conditional variance under GARCH-type models.

Interpretation:

The dominant temporal structure in Bitcoin lies in **second moments**, not first moments.

Finding 3: Asymmetric volatility effects

Using models such as:

- EGARCH
- GJR-GARCH

You observed that:

- Negative shocks often increase volatility more than positive shocks of the same magnitude,
- Volatility responds nonlinearly to innovations.

Interpretation:

Bitcoin volatility exhibits **leverage-like asymmetry**, despite not being a traditional equity.

This is a nontrivial empirical insight.

Finding 4: Heavy-tailed residuals

Even after volatility modeling:

- Residuals deviated from normality,
- Extreme events were underpredicted by Gaussian assumptions.

Interpretation:

Bitcoin returns are better described by **fat-tailed innovation distributions**, reflecting market fragility.

Finding 5: Model adequacy is conditional and regime-dependent

Diagnostics showed:

- Good fit during stable periods,
- Breakdown during extreme market regimes.

Interpretation:

Parameter stability is questionable during structural breaks, which is consistent with crypto market evolution.

3. What these findings mean statistically (this is key)

Your results imply that:

1. **Reduced-form models are informative but incomplete**
 - GARCH captures persistence,
 - But does not explain *why* volatility changes.
2. **Latent state dynamics dominate**
 - Volatility is unobserved,
 - Returns are noisy measurements of that latent process.
3. **Inference difficulty increases with realism**
 - Asymmetry and heavy tails improve fit,
 - But increase estimation instability.

This naturally motivates **state-space and POMP frameworks**.

4. How to summarize your project in one coherent paragraph

Here is a polished explanation you can use verbatim:

My project found that Bitcoin returns exhibit minimal linear predictability in the mean but extremely strong and persistent volatility clustering. GARCH-type models capture this persistence, with asymmetric variants revealing nonlinear responses to shocks and heavy-tailed residuals. While these models fit well during stable regimes, diagnostics deteriorate during extreme volatility, suggesting latent state dynamics and parameter instability that motivate more mechanistic, state-space approaches.

5. How this sets up a natural bridge to Ionides' work

Your findings show:

- Latent volatility,
- Nonlinearity,
- Heavy-tailed noise,
- Parameter instability.

These are **exactly the conditions** where:

- Particle filters are needed,
- Likelihoods become intractable,
- MPIF becomes valuable.

So your project is not “separate” from his work — it is a **reduced-form entry point into the same inferential problem.**

6. One-sentence takeaway (memorize)

The main finding of my Bitcoin project is that virtually all predictable structure lies in latent volatility rather than returns themselves, and that modeling this structure exposes the limits of standard likelihood-based methods under nonlinearity and regime shifts.

Gemini (below):

Executive Summary

The report analyzes Bitcoin's daily closing prices from **September 2014 to July 2025** to understand price volatility and predict future movements. The data reveals that Bitcoin is highly volatile, with prices ranging from \$500 to over \$117,000.

The central challenge identified in the analysis is "**volatility clustering**"—periods where high volatility is followed by more high volatility, and low by low. Because standard models assume constant variance, the analysis moves from basic ARIMA modeling to advanced GARCH variations to better capture Bitcoin's erratic behavior. The ultimate conclusion is that the **EGARCH model** provides the best fit, revealing that Bitcoin reacts differently to "news" (shocks) than traditional stock markets.

Model Analysis & Findings

Here is an explanation of the findings for each specific model utilized in the report:

1. ARIMA (Auto-Regressive Integrated Moving Average)

- **Purpose:** Used to model the "mean" (average) price behavior and linear trends.
- **Configuration:** The best fit was determined to be **ARIMA(2,0,0)** on log-transformed data.
- **Findings:** The model successfully identified that current prices depend on the previous two days' returns.
- **Limitations:** The ARIMA model assumes constant variance (homoscedasticity). However, the residuals (errors) showed significant "volatility clustering," meaning the variance changed over time. This proved ARIMA was insufficient on its own for risk management, necessitating the use of GARCH models.

2. Standard GARCH(1,1) - Generalized Autoregressive Conditional Heteroskedasticity

- **Purpose:** To model the changing volatility (conditional variance) that ARIMA missed.
- **Findings:**
 - **Persistence:** The model confirmed that volatility is highly persistent (a "Beta1" value of 0.92), meaning it takes a long time for the market to calm down after a shock.
 - **Volatility Clustering:** It successfully identified that high volatility days cluster together.
- **Critical Failures:**
 - **Leverage Effect:** It failed the "Sign Bias Test," meaning it could not account for the fact that positive and negative news impact volatility differently.
 - **Model Fit:** It failed the Ljung-Box test, indicating there were still patterns in the data the model didn't capture. The parameters were found to be unstable.

3. EGARCH (Exponential GARCH)

- **Purpose:** Designed to capture "asymmetric" effects (how prices react differently to good vs. bad news) using a logarithmic variance equation.
- **Key Findings (The "Inverse Leverage Effect"):**
 - **The Winner:** This was the **best-fitting model** with the lowest AIC (Akaike Information Criterion) score.
 - **Asymmetry:** The Gamma coefficient was **positive (0.253)**. In traditional stock markets, this number is usually negative (meaning crashes cause panic/volatility). For Bitcoin, the positive value suggests an **"Inverse Leverage Effect"**—**positive news (price jumps) causes more volatility than negative news.**
 - **Inverse Leverage Effect (Very important):** When an asset's volatility **increases following positive price shocks instead of negative ones.** Goes against the leverage effect which is common in stocks, where negative returns lead to higher volatility.
 - **Reliability:** It passed the Sign Bias Test and successfully removed ARCH effects from the residuals.

4. GJR-GARCH - Glosten-Jagannathan-Runkle

- **Purpose:** Another model designed to handle asymmetry, specifically looking at how negative shocks might increase volatility more than positive ones.
- **Findings:**

- The model captured volatility clustering and some leverage effects.
- However, the leverage term was weak/neutral (0.054), and the model had a higher AIC score than the EGARCH model.
- While better than the standard GARCH, it was not as precise as the EGARCH model for this specific dataset.

Final Conclusion

The report concludes that **EGARCH** is the superior model for Bitcoin analysis. It highlights a unique economic characteristic of Bitcoin: unlike traditional equities where fear of loss drives volatility, Bitcoin volatility is driven significantly by **speculative excitement (positive shocks)**, seen in the **Inverse Leverage Effect** where **positive shocks increase the volatility in comparison to negative shocks goes against the original Leverage effect which is common for stocks.**